

17th Feb 2026 | DA6360

- (contd.) $\Rightarrow$ GD
 $\Rightarrow$ GD on $\min \frac{1}{2} x^2$

$$x_{k+1} = x_k - \alpha x_k \quad [x_{k+1} = x_k - \alpha f'(x_k)]$$

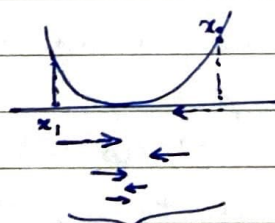
- What is the allowable range of α ?
any $x_0 \in \mathbb{R}$, reach $x_* = 0$ (for convergence)

Ratio Test for convergence

iterates from GD is a seq. so, ratio test applies for GD.

$$|1 - \alpha| < 2, \quad -1 < 1 - \alpha < 1$$
$$0 < \alpha < 2$$

For α btw. $(0, 1)$
 α btw. $(1, 2)$



$\alpha > 2 \Rightarrow$ diverge

oscillate & then converge

- $\min. ax^2 + bx + c, \quad a > 0$
can use derive η for GD.

$$x_{k+1} = x_k - \alpha (2ax_k + b)$$

$$x_* = -\frac{b}{2a}$$

$$(x_{k+1} - x_*) = (x_k - x_*) - \alpha (2a(x_k - x_* + x_*) + b)$$

$$x_{k+1} - x_* = x_k - x_* - \alpha (2a(x_k - x_*) + \underbrace{2ax_* + b}_0)$$

$$x_{k+1} - x_* = x_k - x_* - \alpha 2a (x_k - x_*)$$

$$x_{k+1} - x_* = (1 - \alpha 2a)(x_k - x_*)$$

$$e_k \triangleq x_k - x_*$$

$$e_{k+1} = (1 - \alpha 2a)e_k$$

$$|1 - \alpha 2a| < 1 \Rightarrow -1 < 1 - \alpha 2a < 1$$

$$\alpha \in \left(0, \frac{1}{a}\right)$$

* - for a bag of problems P

$$\left\{ \min f(x), f(x) = ax^2 + bx + c, \quad \begin{array}{l} a > 0 \\ b, c \in \mathbb{R} \end{array} \right.$$

$$\text{GD with } \alpha \in \left(0, \frac{1}{a_{\max}}\right) \}$$

how to identify P ? by particular loss funcⁿ, data distribuⁿ.

- Why does GD work?

taylor
series
approximaⁿ

$$\begin{aligned} f(x_{k+1}) &= f(x_k - \alpha f'(x_k)) \\ &\approx f(x_k) + f'(x_k)(x_k - \alpha f'(x_k) - x_k) \end{aligned}$$

$$\left[f(x) \approx f(\hat{x}) + f'(\hat{x})(x - \hat{x}) \right]$$

$$\begin{aligned} &\approx f(x_k) + f'(x_k)(-\alpha f'(x_k)) \\ &= f(x_k) - \alpha \underbrace{\left(f'(x_k)\right)^2}_{+ve} \end{aligned}$$

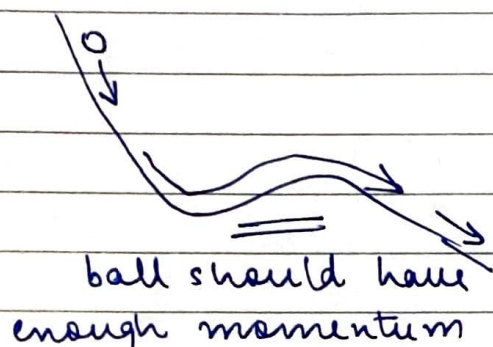
(the design for our algo. was fixed before seeing P)

this is why we hope it works

$$f(x_{k+1}) \leq f(x_k)$$

to check why not exact, pick $f(x) = |x|$, $x_k = 0.05$, $\alpha = 0.1$

⇒ GD with Heavy Ball Momentum (different from Nesterov)



ball should have enough momentum

to cross this part (escape this)

$$x_k - x_{k-1} = -\alpha f'(x_{k-1}) + \beta(x_{k-1} - x_{k-2})$$

$$x_{k+1} = x_k - \alpha f'(x_k) + \beta(-\alpha f'(x_{k-1}) + \beta(x_{k-1} - x_{k-2}))$$

this part can go recursively.

$$= x_k - \alpha f'(x_k) - \alpha \beta f'(x_{k-1}) - \alpha \beta^2 f'(x_{k-2}) \dots$$

decaying memory of past gradients $\beta \in (0, 1)$

$$= x_k - \alpha (f'(x_k) + \beta f'(x_{k-1}) + \beta^2 f'(x_{k-2}) \dots)$$

alternate way of writing

$$\begin{cases} m_k = f'(x_k) + \beta m_{k-1} \\ x_{k+1} = x_k - \alpha m_k \end{cases} \quad \left(\text{unfurl this recursion} \right)$$

⇒ Newton Method $0 \Rightarrow f(x_k), 1 \Rightarrow f'(x_k), 2 \Rightarrow f''(x_k)$
 2nd order $\Rightarrow f(x) \approx f(x_k) + \underbrace{f'(x_k)}_{L^1} (x - x_k) + \frac{f''(x_k)}{2!} (x - x_k)^2 = \hat{f}(x)$
 (x_k is an k^{th} iterate of some algo.)

- let me try to minimize $\hat{f}(x)$ instead of $f(x)$.

$$(\hat{x}_k - x_k) = - \frac{f'(x_k)}{2 f''(x_k)} = - \frac{f'(x_k)}{f''(x_k)}$$

exact solⁿ for $\hat{f}(x)$ we are finding exact solⁿ for approx. quadratic.

but $\hat{f}(x)$ is an approximation to $f(x)$.

this is exactly quadratic

this might take a few steps.

this may not be quadratic

this converges in 1 step.

- Newton method:

$$\left| x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} \right| \begin{array}{l} \leftarrow \text{slope} \\ \leftarrow \text{slope of slope (}\Delta \text{ of slope)} \end{array}$$

$|x| \Rightarrow x_{k+1} = x_k - \frac{\text{sign } x_k}{0}$ this means that the funcⁿ is never going to curve back.

Ex. Compare Newton method vs GD on min.

true min $ax^2 + bx + c$

$$x_{k+1} = x_k - \eta(2ax_k + b)$$

$$\frac{-b}{2a}$$

$$x_{k+1} = x_k - \frac{2ax_k + b}{2a}$$

$$|1 - 2a\eta| < 1$$

$$x_{k+1} = \frac{-b}{2a}$$

$$0 < \eta < \frac{1}{a}$$

- $\min ax^2 + bx^4 \quad (b > 0)$

Newton method

$$x_{k+1} = x_k - \frac{2ax_k + 4bx_k^3}{2a + 12bx_k^2}$$

$$= \left(1 - \frac{2a + 4bx_k^2}{2a + 12bx_k^2} \right) x_k$$

$$x_{k+1} = x_k \frac{GD}{G} = x_k \frac{2ax_k + 4bx_k^3}{2a + 12bx_k^2}$$

$$= (1 - \alpha (2a + 4bx_k^2)) x_k$$

$\alpha, x_0, \dots$

for $|x_k|$ large

$$x_{k+1} \approx \left(1 - \frac{1}{3} \right) x_k$$

$$x_{k+1} \approx \frac{2}{3} x_k$$

for $x_k \ll 1$

$$\left(\frac{2a}{x_k^2} \right) \approx 1$$

- Newton method on $\min x^4, \min^{4/3}$